PassFrame tuition

Volume & 3D Shapes

Year 5 Accelerated — 11+ Lesson Worksheet (PassFrame Tuition)

Worked examples — tutor talks through these first

Easy: What is the volume of a cuboid measuring 4 cm × 3 cm × 2 cm?
Method & answer: Volume = length × width × height = 4 × 3 × 2 = 24 cm³.

Standard: How many faces, edges and vertices does a cube have?
Method & answer: A cube has 6 faces, 12 edges and 8 vertices.

Advanced: A tank measures 20cm × 15cm × 10cm. How many litres of water fill it completely? (1000 cm³ = 1 litre)
Method & answer: Volume = 20 × 15 × 10 = 3000 cm³ = 3000 ÷ 1000 = 3 litres.

Easy / Standard — *work through together, answer 5 in the lesson* ⏱ *15 min (1 min per question)*

1. What is the volume of a cuboid measuring 5cm × 2cm × 3cm?

2. What is the volume of a cube with sides of 4cm?

3. Which of these has a volume of 36cm³? A) 6×3×2 B) 9×2×2 C) both A and B D) neither

4. How many edges does a cube have?

5. How many vertices does a cube have?

6. What is the volume of a cuboid measuring 6cm × 3cm × 2cm?

7. Which 3D shape has exactly 2 flat faces (both circles) and 1 curved face? A) cone B) cylinder C) sphere D) cuboid

8. How many faces does a cylinder have (flat and curved together)?

9. Select ALL of these that are true properties of a triangular prism: A) 5 faces B) 9 edges C) 6 vertices D) 2 triangular faces E) all faces are rectangles

10. What is the volume of a cube with sides of 5cm?

11. How many vertices does a square-based pyramid have?

12. What is the volume of a cuboid measuring 8cm × 4cm × 1cm?

13. How many edges does a triangular prism have?

14. How many vertices does a sphere have?

15. What is the volume of a cube with sides of 3cm?

Standard — *work through together, answer 5 in the lesson* ⏱ *15 min (1 min per question)*

1. A cuboid has a volume of 60cm³. Its length is 5cm and width is 4cm. What is its height?

2. A square-based pyramid has 5 faces, 8 edges and 5 vertices. What is the total if you add all three?

6. How many edges does a cone have?

7. A fish tank measures 30cm × 20cm × 25cm. What is its volume in litres?

11. Cuboid A measures 4×3×2cm; cuboid B measures 3×3×3cm. Which has the greater volume?

12. A cube's 6 faces are identical squares, each with an area of 16cm². What is the length of one edge?

3. Convert 2000 cm^3 to litres (1000 cm^3 = 1 litre).

4. A cube has a volume of 125cm^3. What is the length of one side? A) 5cm B) 15cm C) 25cm D) 125cm

5. A cuboid measures 10cm × 5cm × 4cm. What is its volume?

8. Select ALL of these statements that are TRUE about the net '6 squares arranged in a cross shape': A) it folds to make a cube B) it has 6 faces C) it folds to make a triangular prism D) each square becomes a face of the cube E) it has 8 faces

9. A cuboid's volume is 96cm^3. Its length is 8cm and height is 3cm. What is its width?

10. How many faces does a square-based pyramid have?

13. Convert 3.5 litres to cm^3.

14. A cuboid has dimensions 7cm × 2cm × 5cm. What is its volume?

15. How many vertices does a triangular prism have?

Standard / Hard — *work through together, answer 5 in the lesson* *15 min (1 min per question)*

1. A tank measures 40cm × 30cm × 25cm and is half full of water. How many litres of water are in the tank?

2. A cuboid box has a volume of 240cm^3, with length 10cm and width 6cm. What is its height, and how many of these boxes fit exactly into a larger box of volume 960cm^3? A) 2cm; 4 boxes B) 4cm; 4 boxes C) 4cm; 2 boxes D) 2cm; 2 boxes

3. A cube-shaped box has a volume of 512cm^3. What is the length of one edge, and what is the box's total surface area?

4. A pool is 12m long, 6m wide and 1.5m deep throughout. How many cubic metres of water fill it, and how many litres is this (1m^3 = 1000 litres)?

5. A cuboid has a square base with side 5cm and a volume of 150cm^3. What is its height?

6. Select ALL of these statements that are TRUE about a 512cm^3 cube-shaped box: A) its edge is 8cm B) its edge is 512cm C) its surface area is 384cm^2 D) it has 8 vertices E) its surface area is 512cm^2

7. A tank measures 50cm × 25cm × 30cm and is filled to a depth of 20cm. How many litres of water does it contain?

8. A cuboid's length, width and height are in the ratio 3:2:1. If the height is 4cm, what is the volume?

9. A cube has a volume of 1000cm^3. It is melted down and recast into a cuboid with a base of 10cm × 5cm. What is the height of the new cuboid?

10. A tank measures 2m × 1.5m × 1m and is filled at 500 litres per minute. How long does it take to fill completely? (1m^3 = 1000 litres)

11. A 20cm × 15cm × 10cm box is packed with 5cm cubes, with no gaps or overhang. How many small cubes fit inside?

12. A cuboid measures 9cm × 8cm × 5cm. It is cut exactly in half, dividing its 5cm height into two equal layers. What are the dimensions of each half?

13. A cuboid has a volume of 360cm^3. Its length is twice its width, and its height is 5cm. What are its length and width?

14. A 25cm × 20cm × 16cm container, completely full of water, is poured into empty cube-shaped bottles of side 10cm. How many bottles can be completely filled, and how much water is left over?

15. A cuboid box has three different face areas: 20cm^2, 15cm^2 and 12cm^2. What is the volume of the box?